Agenda

Optimization Hive Query

(a) Text vs Parquet vs Avro

(b) Partition and Bucketing

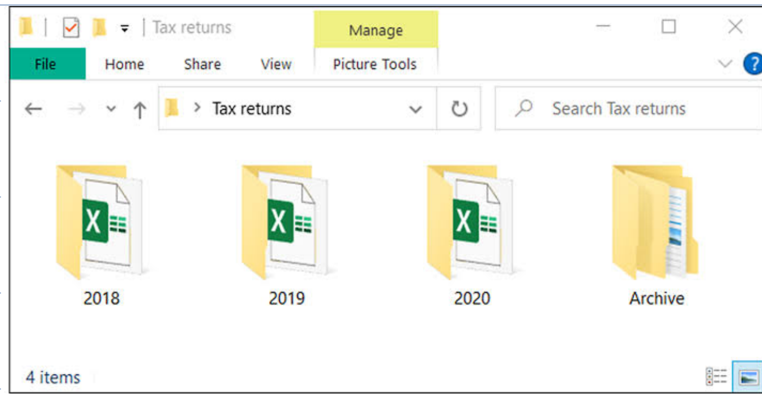
(c) Different Processing Engine

Introduction to Cloud

Architecture – AWS Redshift

Demo Redshift

Optimization - Hive Query



Post Office Dataset

Contains information about Post Offices in India

→ Store this in HDFS, create a normal table

1. Push this csv file to your HDFS Directory
2. You create a schema on Hive with external location pointing to your location
3. You can count number of PO in HARYANA

select count (*) from table
where state = 'HARYANA';

Partition

Look at the column based on which your data should be organized and creates dynamic folder for all unique value in the column

↳ create a partitioned table

→ metastore will store all partition location

Static partition

Majority of times we are interested in subset of data

It does not make logical sense to store all values again

→ majority of times we are interested in static partition during dev phase

Processing Engine

Map Reduce → 2016

Tez

Spark [future class]

Bucketing → Hash Function → postal code (Int)

Logic of splitting data using a Hash function into multiple manageable data storage

Postal	state	city
110092		
600075		
500181		
000 002		

→ 4 buckets % 4

0	11 0092	1	500181
2	000 002	3	600075

This usually is done along with partitioning to manage extremely large data sets

It becomes extremely important for us to choose appropriate column and key value

→ Query on top of your key will be faster

Introduction to Cloud

No Furnishing

Legacy servers

Fully Furnished

Cloud — take care of your logic

Data Engineering

Lot of servers with compute, storage, memory and network are needed

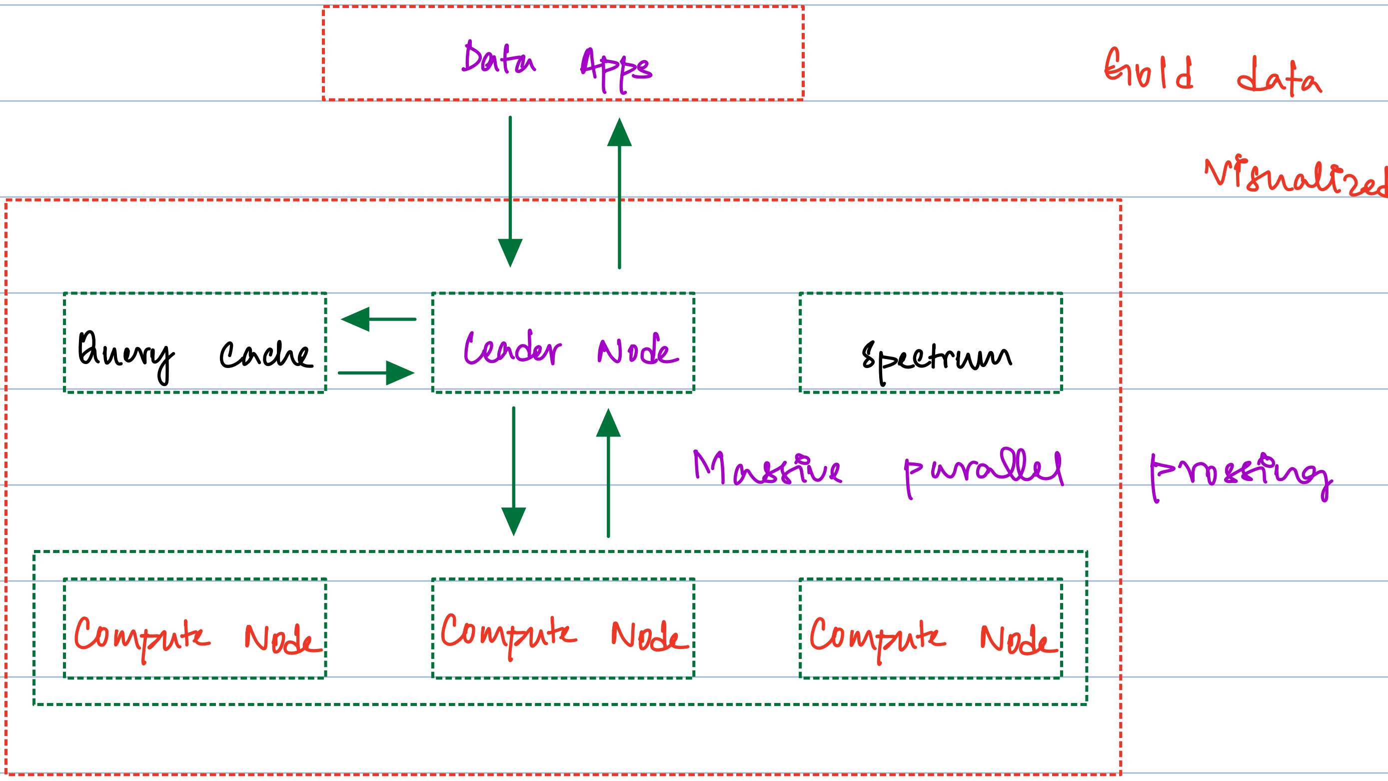
Hadoop → 2 to 3 weeks by yourself

AWS

① IAM — Identity manager

② Creating, Practicing and destroying services immediately

Architecture of AWS Redshift



Data Apps

Client that interacts with the Redshift cluster

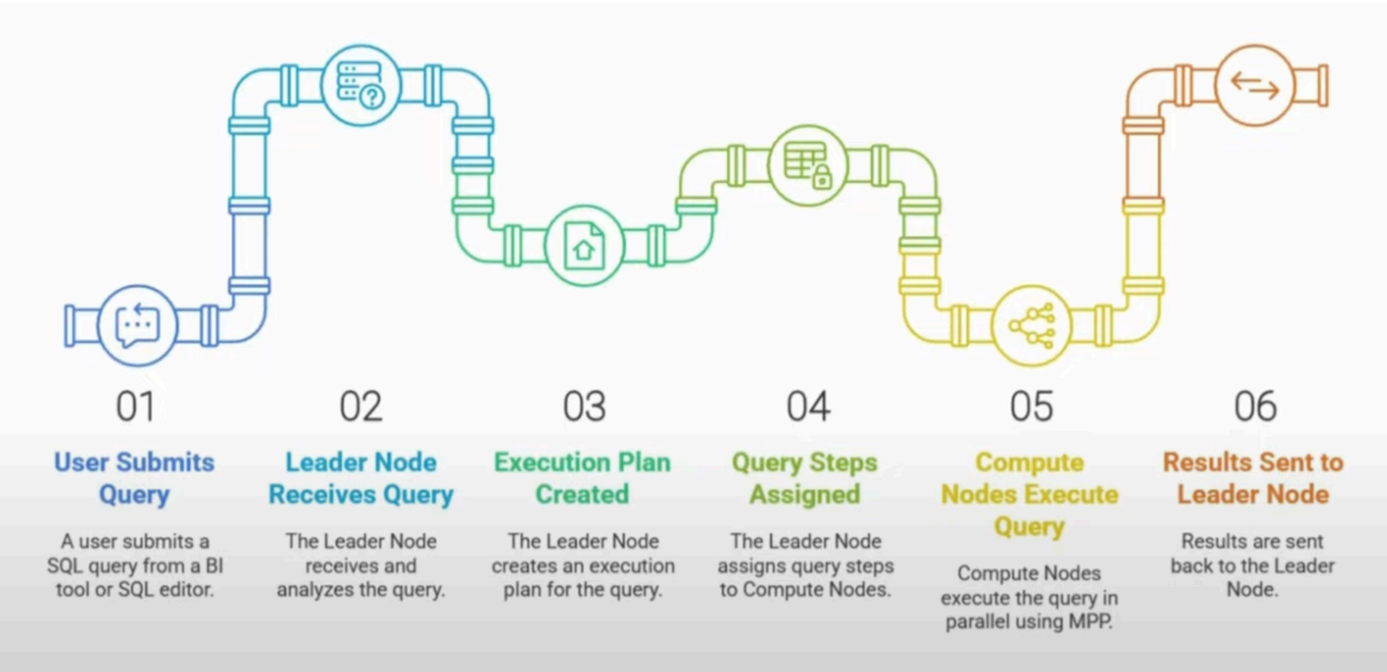
Leader Node

Similar to the master node, handles the request and analyses, plans and optimizes the query

Compute Node

Fetches the data from storage using MPP

How does a Query Execute



Types of nodes available

1. DC2 : Dense compute Node
2. RA3 : Redshift A series nodes

Spectrum Service

You manage your data in your S3 Buckets & will create FaaS every time a request comes in

Demo Redshift

1. Load data into S3
2. You create a AWS Redshift serverless
3. You create table
4. Load the data

Optime our Queries in AWS Redshift

1. Distribution key
2. Sort

Next class

1. Demo
2. Optimizations in Redshift

Announcement

→ Additional class already informed